

Industrial Internship Report on

Machine Learning Based Predictive Analytics: Quality Prediction in a Mining Flotation Process & Smart City Traffic Forecasting

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Executive Summary

This report presents the work completed during the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

The internship was focused on two data science / machine learning problem statements provided by UCT, completed along with this report within the internship period.

The first project builds a machine learning pipeline to predict the percentage of Silica Concentrate in the output of an iron-ore froth flotation plant, so that silica impurity levels can be controlled without relying solely on slow, periodic laboratory assays.

The second project builds a machine learning framework to forecast smart-city traffic volumes at different road junctions using historical time-series traffic data, in order to support better urban planning and real-time traffic management.

This internship gave a valuable opportunity to work on real industrial and civic data problems end-to-end — from data cleaning and exploratory analysis to model building, evaluation, and deployment-ready packaging — and was an overall great learning experience.

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Preface

This report summarizes six weeks of Industrial Internship work carried out under upskill Campus (USC) and The IoT Academy, in association with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

An internship of this kind is an important step in career development: it moves learning out of a purely academic setting and into the context of real business and engineering constraints — messy data, ambiguous requirements, time limits, and the need to justify design choices with evidence rather than assumption.

During the internship, two problem statements were assigned: (1) predicting silica impurity in the concentrate of an iron-ore flotation plant, and (2) forecasting traffic volumes at smart-city road junctions. Both projects followed the same overall program structure — problem framing, data preprocessing, exploratory data analysis, model development, and evaluation — delivered against a fixed timeline set by USC/UCT.

The internship provided hands-on exposure to industrial and civic data problems, the opportunity to design and implement end-to-end machine learning solutions for them, and mentorship from the UCT team throughout. It was an overall valuable and enjoyable experience.

Sincere thanks to the mentors and the UCT/USC team for their guidance and support throughout the internship, and to peers for their collaboration.

To juniors and future interns: start with the data before the model, keep the EDA thorough, and don't be afraid of the simplest baseline — it tells you how hard the real problem actually is.

Introduction

About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a company established in 2013, working in the Digital Transformation domain and providing industrial solutions with a prime focus on sustainability and Return on Investment (RoI).

To develop its products and solutions, UCT leverages cutting-edge technologies such as the Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and front-end engineering.

UCT's key platforms and solution areas include the UCT IoT Platform ("UCT Insight") for rapid IoT application deployment and analytics; the Smart Factory Platform ("Factory Watch") for production and asset monitoring, OEE, and predictive maintenance; LoRaWAN-based solutions for Agritech, smart cities, and industrial monitoring; and predictive maintenance solutions combining embedded systems, Industrial IoT, and Machine Learning to estimate the remaining useful life of production machinery.

About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in an affordable, scalable, and measurable way.

The IoT Academy

The IoT Academy is the EdTech division of UCT, running long-term executive certification programs in collaboration with EICT Academy and IIT Kanpur, IIT Roorkee, and IIT Guwahati across multiple domains.

Objectives of this Internship program

The objectives of this internship program were to:

- Get practical experience of working in industry on real data science problems.
- Solve real-world problems using structured data science and machine learning workflows.
- Improve job prospects through demonstrable, evaluated project work.
- Deepen understanding of machine learning and its industrial and civic applications.
- Build personal and professional skills such as communication, documentation, and problem solving.

Reference

[1] MiningProcess_Flotation_Plant_Database.csv — iron-ore flotation plant sensor dataset (project 1).

[2] Smart city traffic dataset and project09.ipynb notebook (project 2).

[3] scikit-learn, XGBoost, and CatBoost documentation.

Glossary

Terms	Acronym / Meaning
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
EDA	Exploratory Data Analysis
ML	Machine Learning
XGBoost	Extreme Gradient Boosting

Problem Statement

Two independent problem statements were assigned during this internship, both framed as supervised regression / forecasting tasks on real sensor and time-series data.

Project 1: Quality Prediction in a Mining Process

Controlling silica impurity in iron-ore concentrate is vital for maximizing product quality and minimizing environmental and operational costs in a froth flotation plant. Silica content is traditionally confirmed by laboratory assay, which is slow and cannot support real-time process control. The task was to build a machine learning model that predicts the percentage of Silica Concentrate (% Silica Concentrate) directly from real-time plant sensor data, including % Iron Feed, % Silica Feed, ore pulp pH, ore pulp density, and reagent/air-flow readings from the flotation columns.

Project 2: Smart City Traffic Forecasting

Traffic congestion in urban areas is a growing problem, and city planners need reliable forecasts of vehicle volumes at different junctions to plan interventions and manage traffic in real time. The task was to build a machine learning framework that forecasts future traffic volumes at multiple city junctions using historical, time-stamped traffic count data, capturing hourly, daily, and location-based patterns.

Existing and Proposed solution

Project 1: Mining Process Quality Prediction

Existing approach: Silica concentrate percentage is determined mainly through periodic laboratory sampling and assay, which introduces a significant time lag (often hours) between when a sample is drawn and when the result is known, limiting the plant's ability to react quickly to quality drift.

Proposed solution: A data-driven regression pipeline was built to predict % Silica Concentrate directly from existing real-time sensor streams, removing the lab-assay time lag. The data was cleaned (comma-decimal strings converted to floats, duplicates dropped, missing values handled) and enriched with time-based features (Year, Month, Day, Hour, Minute, DayOfWeek) extracted from the timestamp column. Features were standardized with StandardScaler, and five regression algorithms — Linear Regression, Decision Tree, Random Forest, Extra Trees, and Gradient Boosting — were trained on an 80/20 train-test split and compared.

Value addition: Near-instant, sensor-driven silica prediction (rather than hours-delayed lab results) that plant operators can use to adjust reagent dosing and air-flow settings proactively, improving concentrate quality and reducing waste.

Project 2: Smart City Traffic Forecasting

Existing approach: Traffic management in many cities still relies on fixed-time signal plans or manual observation rather than data-driven forecasts, which does not adapt well to real variation in traffic volume across time and location.

Proposed solution: A forecasting pipeline was built using historical junction-level traffic counts. EDA uncovered hourly, daily, and monthly cyclical patterns, along with strong spatial variation across junctions. Temporal and location features (hour of day, day of week, junction ID) were engineered, and four models — Linear Regression, Random Forest, XGBoost, and CatBoost — were trained and compared using MAE, RMSE, and R^2 .

Value addition: A forecasting model that captures both time-of-day/day-of-week cycles and per-junction differences, giving city planners a quantitative basis for proactive traffic management rather than reactive, fixed-schedule control.

Code submission and Report submission (Github link)

<https://github.com/Maitri05patel/upskillcampus>

Proposed Design / Model

Both projects followed the same overall data-science pipeline design: raw data ingestion → cleaning → feature engineering → exploratory analysis → model training and comparison → selection and serialization of the best model. DS/ML-specific detail for each stage is given below.

High Level Diagram

Figure 1: End-to-end pipeline — Raw sensor / traffic data → Data Cleaning → Feature Engineering (datetime features) → EDA (distributions, correlations, boxplots) → Train/Test Split → Feature Scaling → Model Training (multiple algorithms) → Model Comparison → Best Model Serialization.

Low Level Diagram

Project 1 — Mining Process

- Type conversion: comma-decimal strings → floats.
- Datetime engineering: date → Year, Month, Day, Hour, Minute, DayOfWeek; original date column dropped.
- Duplicate and missing-value handling, with a missing-value heatmap for verification.
- EDA: histograms, boxplots, correlation heatmap, and scatter/pair plots (e.g. Ore Pulp Density vs. % Silica Concentrate, % Iron Feed vs. % Silica Concentrate).
- 80/20 train-test split; StandardScaler applied to features.
- Five regressors trained: Linear Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting.
- Best model (Extra Trees Regressor) serialized as best_regression_model.pkl.

Project 2 — Smart City Traffic Forecasting

- EDA on hourly, daily, and monthly traffic trends, and per-junction traffic comparison.
- Feature correlation analysis to identify the most impactful predictors.
- Feature engineering: hour of day, day of week, junction ID extracted as model inputs.
- Four regressors trained: Linear Regression, Random Forest, XGBoost, CatBoost.
- Model outputs compared against ground truth (Actual vs. Predicted plots) and feature importance analyzed.
- Final predictions exported to Predictions/submission.csv.

Interfaces

Both pipelines are notebook-driven (Jupyter). Data flows from CSV files on disk, through pandas-based cleaning and feature engineering, into scikit-learn / XGBoost / CatBoost estimators; trained artifacts (best_regression_model.pkl for Project 1) and plots are written back to disk (Results/Plots and EDA/Plots directories respectively) for downstream use or deployment.

Performance Test

This section quantifies why the chosen models are suitable for real deployment rather than being purely academic exercises. The main constraints considered were prediction accuracy (MAE, RMSE, R^2) and training time, since a plant or city control loop needs both an accurate and a practically trainable/updatable model.

Test Plan / Test Cases

For each project, all candidate models were trained on the same 80% training split and evaluated on the same held-out 20% test split, using identical preprocessing (scaling, feature engineering), so that MAE, RMSE, R^2 , and training time are directly comparable across models.

Test Procedure

Models were fit on the training set, predictions were generated on the held-out test set, and MAE, RMSE, and R^2 were computed against the true target values (% Silica Concentrate for Project 1; traffic volume for Project 2). Training time was also recorded for each model to evaluate the accuracy/cost trade-off.

Performance Outcome

Project 1: Mining Process — Model Comparison

Model	MAE	RMSE	R^2 Score	Training Time
Extra Trees	0.0016	0.0073	0.9999	6.5 mins
Random Forest	0.0030	0.0204	0.9997	37 mins
Decision Tree	0.0020	0.0370	0.9989	1 min
Gradient Boosting	0.3665	0.4972	0.8042	13 mins
Linear Regression	0.4824	0.6266	0.6890	2.3 secs

The Extra Trees Regressor drastically outperformed the other models, achieving an almost perfect R^2 score (0.9999) and the lowest MAE/RMSE, while training considerably faster than Random Forest (6.5 vs. 37 minutes). Random Forest was also highly accurate but far slower to train. This indicates that ensemble tree-based models capture the complex, non-linear relationships in the plant's sensor data extremely well. The best model was serialized as `best_regression_model.pkl` for future inference. A constraint to note for deployment is training/retraining time; Extra Trees offers the best accuracy-to-time trade-off among the ensemble methods tested.

Project 2: Smart City Traffic — Model Comparison

Model	MAE	RMSE	R^2 Score
XGBoost	5.013	7.880	0.9162
Random Forest	4.939	8.423	0.9043
CatBoost	7.555	11.920	0.8083

Model	MAE	RMSE	R ² Score
Linear Regression	11.503	16.584	0.6290

XGBoost achieved the best overall performance, with the highest R² (0.9162) and lowest RMSE (7.880), explaining 91.6% of the variance in traffic volume and handling outliers effectively. Random Forest was a close alternative, with a marginally lower MAE (4.939) but a slightly higher RMSE. Both ensemble methods, along with CatBoost, clearly outperformed the Linear Regression baseline, confirming that traffic patterns are highly non-linear. A constraint for this use case is inference latency for real-time traffic management; tree-ensemble models such as XGBoost offer fast inference suitable for near-real-time forecasting once trained.

My learnings

This internship reinforced a full, disciplined data-science workflow across two very different domains — industrial process sensing and urban traffic systems — which showed how the same core pipeline (clean → explore → engineer features → model → evaluate → compare) generalizes across problem types.

- Data cleaning is not a formality: locale-specific number formats (comma decimals) and timestamp parsing directly determine whether downstream models can even be trained correctly.
- EDA before modeling consistently paid off — correlation heatmaps and time-based plots pointed directly at the features that ended up mattering most (e.g. sensor readings for silica; hour/day/junction for traffic).
- Comparing multiple algorithms rather than committing to one early is essential: simple baselines (Linear Regression) quantified how non-linear each problem really was, and ensemble tree methods (Extra Trees, Random Forest, XGBoost) consistently won on accuracy.
- Model selection is a trade-off, not just an accuracy ranking — training time and expected retraining frequency matter for real deployment (e.g. Extra Trees vs. Random Forest in Project 1).
- Clear documentation and reproducible pipelines (README, saved metrics, serialized models) make a project usable by others, not just by its original author.

Overall, this experience strengthened both technical machine learning skills and the discipline of communicating results (metrics, plots, trade-offs) to a non-technical audience, both of which will directly support future career growth.

Future work scope

Ideas that could not be completed within the internship timeline but could be pursued in future work:

- Project 1: Deploy the Extra Trees model as a real-time inference service integrated with plant sensor streams, with monitoring for data/model drift and periodic retraining.
- Project 1: Explore feature importance / SHAP analysis to identify which sensors most strongly drive silica concentrate, to guide process control decisions.
- Project 2: Incorporate additional context such as weather, local events, and holidays to improve traffic forecasts.
- Project 2: Extend the model to multi-step-ahead forecasting (e.g., predicting traffic several hours in advance) rather than single-point prediction.
- Both: Package each pipeline into a lightweight API/dashboard for non-technical stakeholders to query predictions directly.